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DATA DESCRIPTOR

Noise mapping and ambient sound recordings of the urban environment in Uganda

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Urban noise pollution is a growing public health concern, particularly in rapidly developing cities where regulatory enforcement is limited. High noise levels have been linked to adverse health effects, including stress, sleep disturbances, and cardiovascular issues. However, data-driven noise monitoring remains scarce in many low-resource settings, limiting efforts to develop effective mitigation strategies. To address this gap, we designed and deployed a large-scale noise data collection pipeline in Kampala and Entebbe, Uganda. Our approach integrates mobile-based tools to capture short audio recordings, categorizations of the type of noise, and sound pressure levels at specific locations. Over the course of the study, we collected 61,821 such annotated noise samples across five divisions in Kampala and four wards in Entebbe. This is to our knowledge the largest existing urban sound dataset, and the first time that the acoustic landscape of a developing-world city has been characterized in such detail. It provides valuable insights into urban noise pollution, forming a foundation for future noise mapping and mitigation efforts.

Background & Summary

Noise pollution is a severe unnoticed threat which continues to spread due to population increase, urbanisation, growth and development associated with sources of noise such as highways, factories and construction, among others. This is particularly a growing issue in the developing world, stemming from rapid urbanization. Studies have shown that continued exposure to environmental noise pollution poses a danger to public health by causing problems like disruption of sleep, increased rate of hypertension and cardiovascular disease, and decline in cognitive performance in children^{1,2}. The varied health effects of noise pollution specifically on children are a great cause for concern and have become the focus of recent research³.

In addition to health, noise pollution can also negatively affect economic activity by disrupting some types of trade and business. In Uganda, there have been cases where real estate owners have not obtained long-term tenants because they are located in extremely noisy places.

Despite the extent of these problems, there are almost no suitable datasets with which to inform policy or plan mitigations, particularly in the developing world. We summarize currently available urban sound datasets below, but note two major shortcomings:

1. Many of existing urban noise datasets are collected as clips from websites that provide free audio samples either scraped from the web or uploaded by users, and hence are not consistently recorded in the field.
2. Urban sound datasets typically include audio recordings, but without precise geographical locations, time of day or sound pressure levels. To build up a comprehensive picture, all of these are necessary.
3. There is a lack of noise datasets collected and curated in the developing world, especially Africa. For the datasets which do exist, the categories of sound collected have limited relevance in this context.

The motivation for this work is therefore to collect noise data within an African city and curate a dataset of noise samples across various relevant categories. To ensure the dataset's usefulness for urban noise monitoring and complaint resolution, we collaborated with city authorities to identify key noise categories of concern.

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The resulting dataset, Urban Noise Uganda 61K, is publicly available on Figshare (<https://doi.org/10.6084/m9.figshare.30168901>)⁴, and has a range of practical applications.

Value of the Data. The value of this dataset lies in its ability to provide valuable insights into urban noise patterns, supporting various applications in noise pollution monitoring and smart city development. Key aspects of its value include:

- The dataset provides a rich collection of urban noise recordings from diverse environments, including five divisions of Kampala and four wards in Entebbe. It is the largest urban sound dataset anywhere in the world and the first time such a dataset has been produced for an African city.
- The dataset is useful for training machine learning models for noise classification, environmental sound analysis, and urban noise pollution monitoring, offering valuable insights for city authorities, researchers, and policy makers.
- Unlike existing datasets such as UrbanSound8K and SONYC-UST, this dataset includes noise measurements, precise geospatial metadata (latitude, longitude, altitude, accuracy), and timestamps, enabling detailed spatiotemporal noise mapping.
- The dataset has been collected using a standardized mobile-based data collection pipeline, ensuring high-quality and consistent audio recordings sampled at 16 kHz.
- This dataset can support the development of AI-driven applications for noise pollution mitigation, real-time noise monitoring, and smart city planning in developing urban environments.

Related work. Environmental sound datasets have become increasingly sophisticated tools for understanding urban acoustics and noise pollution. Early collections primarily focused on limited sound event recognition with basic classification systems, but contemporary datasets have expanded to capture the complex acoustic tapestry of urban environments. Modern environmental sound datasets typically comprise several key components: audio recordings in standardized formats (commonly WAV), contextual metadata about recording conditions, taxonomies of sound categories, and in some cases, decibel measurements or other quantitative metrics of sound intensity. These datasets serve multiple research purposes, from developing automated noise classification systems to informing urban planning decisions and public health initiatives related to noise pollution. The methodological approaches to dataset creation have also evolved, with increasing emphasis on capturing real-world conditions rather than laboratory settings, comprehensive spatial and temporal coverage, and standardized collection protocols to ensure consistency across diverse urban environments. This evolution reflects the growing recognition of urban soundscapes as critical components of environmental quality and public health considerations.

Major Urban Sound Datasets.

- **UrbanSound8K** is a comprehensive collection of 8,732 labeled urban sound samples classified into ten distinct categories including car horns, sirens, and children playing. Developed by⁵ in 2014, these recordings have a maximum duration of 4 seconds, are sampled at rates between 16kHz and 48kHz, and include metadata about recording locations and devices used. This dataset has become an essential benchmark for evaluating audio classification models focused on urban environments^{5,6}.
- **SONYC Urban Sound Tagging (SONYC-UST)** dataset was developed from New York City's acoustic sensor network, containing 3,068 recordings specifically designed for real-world urban noise monitoring. The dataset features annotations for 23 fine-grained sound classes that can be grouped into eight coarse-grained categories, including engines, machinery impacts, powered saws, alert signals, and dog barks. These classifications were developed in consultation with the New York City Department of Environmental Protection, highlighting the dataset's practical applications for urban noise management^{7,8}.
- **DCASE 2019-Task 5** dataset, derived from SONYC recordings, was specifically compiled for the DCASE 2019 challenge on multi-label Urban Sound Tagging. It features recordings from New York City with annotations for 23 urban sound noise labels organized into 8 main categories. All recordings are standardized as 10-second, single-channel, 44.1 kHz, 16-bit wave files⁹.
- **ESC-50** provides a broader environmental sound classification resource with 50 classes spanning diverse sound categories including natural sounds (rain, wind), human sounds (laughing, coughing), urban noises (car horns, sirens), and domestic sounds (clock ticking, door knocking). Unlike datasets focused specifically on urban environments, ESC-50 offers a wider acoustic perspective with recordings collected globally through the Freesound online platform¹⁰.
- **Noise Monitoring Data In India (2011-2018).** This dataset was compiled by the Central Pollution Control Board (CPCB) under the National Ambient Noise Monitoring Network (NANMN) in India. Includes data from 70 monitoring stations across seven major cities: Delhi, Mumbai, Kolkata, Chennai, Bengaluru, Hyderabad, and Lucknow. The dataset contains noise level readings at monthly level of various stations across seven cities in India¹¹.

Classification Standards for Noise Levels. Urban noise mapping and monitoring rely on established classification standards to categorize sound levels and assess their potential impacts on human health and wellbeing. These standards typically divide the acoustic environment into distinct categories based on decibel measurements. A widely adopted classification system categorizes noise levels into five main groups: very quiet (10-25 dB), quiet (25-40 dB), moderate (40-55 dB), loud (55-70 dB), and very loud (>70 dB)¹². These classifications are

No.	Noise (dB)	ODK (dB)	Casella (dB)
0	35	36.76	35.12
1	45	45.40	46.16
2	55	55.98	55.61
3	65	67.24	65.10
4	75	76.87	75.36
5	85	84.02	85.27
6	95	95.95	96.20
7	105	104.85	104.84
8	115	114.90	115.25
9	120	120.41	119.32

Table 1. Comparison of noise levels measured by smartphone (ODK App) and Casella Meter.

particularly important because noise levels above 55 dB on LDEN (day-evening-night equivalent level) are generally considered “high noise levels” that may contribute to adverse health effects.

The 55 dB threshold is significant in noise pollution research and urban planning as it represents a point beyond which prolonged exposure may contribute to stress, sleep disturbances, and other health impacts documented in environmental health literature. These standardized classifications enable researchers and policymakers to communicate noise data consistently across different urban contexts and develop targeted interventions for areas experiencing excessive noise. For urban noise mapping projects like the one conducted in Uganda, these classification systems provide essential frameworks for interpreting raw acoustic data and translating them into meaningful insights for public health and urban planning applications.

Gap in Global South Representation. While datasets from cities like New York and other Global North locations have advanced our understanding of urban acoustics, there remains a significant gap in high-quality environmental sound data from the Global South. This under-representation is particularly concerning for fast-growing urban centers in developing regions where noise pollution often follows different patterns and presents unique challenges compared to cities in developed countries¹³.

The lack of geographically diverse sound datasets creates several limitations for noise monitoring efforts in developing regions. First, existing machine learning models trained predominantly on Global North data may not generalize well to the distinct acoustic landscapes of Global South cities, where different transportation systems, construction practices, and cultural activities shape the soundscape as this is evidenced by research in South Africa on quantifying noise pollution levels^{14,15}. Second, the absence of locally relevant data hampers the development of context-appropriate noise mitigation strategies that could address the specific needs of these communities as seen with research done in Ghana¹⁶. Third, this data gap reinforces existing inequities in environmental monitoring capabilities between developed and developing regions. A good case scenario was work done to establish a Spatiotemporal and Environmental Analysis urban noise pollution dataset in El Mansoura Egypt from May to December 2023 that provided variation of decibels by day and night rather than a more detailed dataset inclusive of sample urban noise sounds to serve as evidence for the noise pollution sources¹⁷.

This representational imbalance in available datasets underscores the importance of initiatives like the Uganda noise mapping project described in this paper. By collecting extensive noise data across multiple urban areas in Uganda, this work begins to address the critical need for more diverse acoustic datasets that capture the varied realities of noise pollution in Global South contexts. Such locally collected and curated datasets are essential for developing more inclusive and globally applicable approaches to urban noise monitoring and mitigation.

Methods

Equipment. Data was collected using 10 Tecno Spark 4 Android mobile phones. This phone model was selected for its reliable storage capacity, long battery life, and overall suitability for field-based data collection.

In order to ensure dependable results for measuring sound pressure levels, the mobile phones were calibrated and correlated against the a class 1 casella CEL-633A1 sound level meter, to ensure a reliable metric for the decibel level measurement Table 1.

Figure 1 summarizes the comparison between smartphone (ODK App) and Casella meter readings across these calibration levels.

The calibration results show strong agreement between the smartphone (ODK App) measurements and the Casella CEL-633A1 sound level meter across the entire range of tested noise levels. The differences observed remain within acceptable error margins, confirming that the smartphone setup provides reliable and accurate decibel measurements. This ensures consistency and validity in the noise data collected during the study.

Data Collection System Architecture. We used the Open Data Kit (ODK) platform and an Android audio recorder app from the Google Playstore. Open Data Kit (<https://getodk.org/>) is a platform that lets you build powerful offline forms to collect data. We used ODK Central to store the collected data and ODK Collect to collect the data. Audio Recorder is an Android application for recording audio.

Our architecture allows data collection via an Android handheld device and is based on open data standards such as XForms and HTTP protocols for data transmission. The system consists of the following components:

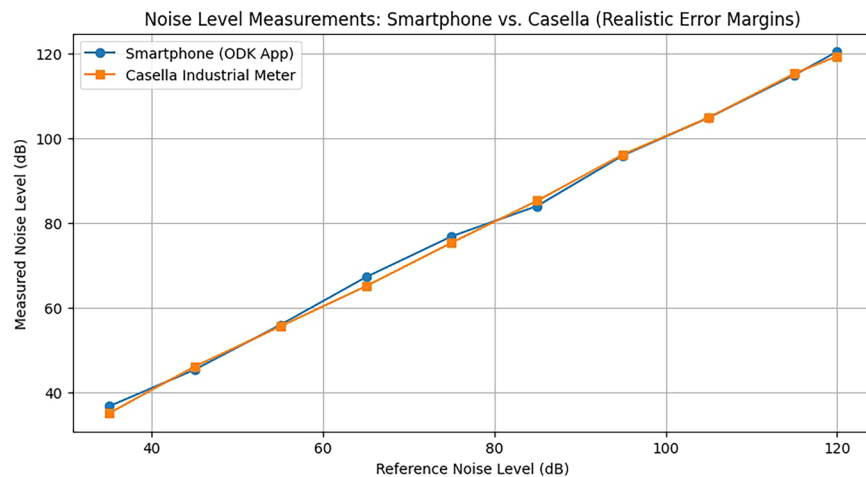


Fig. 1 Calibration of smartphone-based noise measurements against a Class 1 sound level meter. This figure compares measured noise levels obtained using the ODK smartphone application with reference measurements from the Casella CEL-633A1 industrial sound level meter across a controlled range of 35–120 dB. The plot shows strong linear agreement between the two devices, with deviations remaining within acceptable error margins. These results validate the reliability of the smartphone-based data collection setup used in the study.

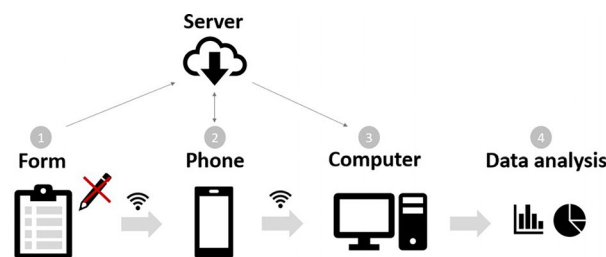


Fig. 2 Architecture of the mobile-based environmental noise data collection system. This schematic illustrates the components and data flow within the noise collection pipeline. The system is built around an ODK XForm deployed on Android phones, which capture audio recordings, noise measurements, and GPS data. Data are uploaded to ODK Central (server) and subsequently accessed via computer for cleaning, analysis, and quality control. The architecture supports offline-first operation suitable for field conditions in Kampala and Entebbe.

- **Form:** Used to capture noise audio, noise measurements, and location data.
- **Phone:** The mobile device where the ODK Collect application is installed and used to fill forms.
- **Server:** ODK Central, used to store and manage collected data.
- **Computer:** Some data uploads can be made directly to the server via a web application.
- **Data Analysis:** Involves data cleaning and preparation for further investigations.

This architecture is summarized schematically in Fig. 2, which illustrates the interaction between the form, phone, server, and data analysis components in the pipeline.

Data collection. The data collection team comprised twenty research assistants assigned across the five divisions of Kampala and Entebbe Cities. Over a five-week period, they were tasked with collecting environmental audio data, targeting a wide range of noise categories. The team operated on foot and via boda-bodas (motorcycle taxis), allowing for greater mobility and coverage across both densely populated and quieter areas. In locations with minimal noise, recordings were still made and categorized under “silence” or “other” to ensure a comprehensive soundscape profile.

Prior to deployment, the team was equipped with Android smartphones pre-installed and configured with the ODK Collect app, an audio recorder application, and the necessary QR codes to access the data collection forms. The ODK Central server was used for centralized form design, deployment, and ongoing data monitoring. We designed a customized ODK form tailored to capture metadata along with each audio recording, including location, time, and a manually assigned sound category. Team members were trained in using the system, with particular emphasis on consistent sound categorization and proper handling of recording devices.

Each recording captured was at least 10 seconds in duration, and sound pressure levels were measured using the audio recording app. Despite limitations in nighttime data collection due to safety concerns, some data collectors, particularly those already operating at night as motorcycle taxi drivers, contributed valuable nighttime

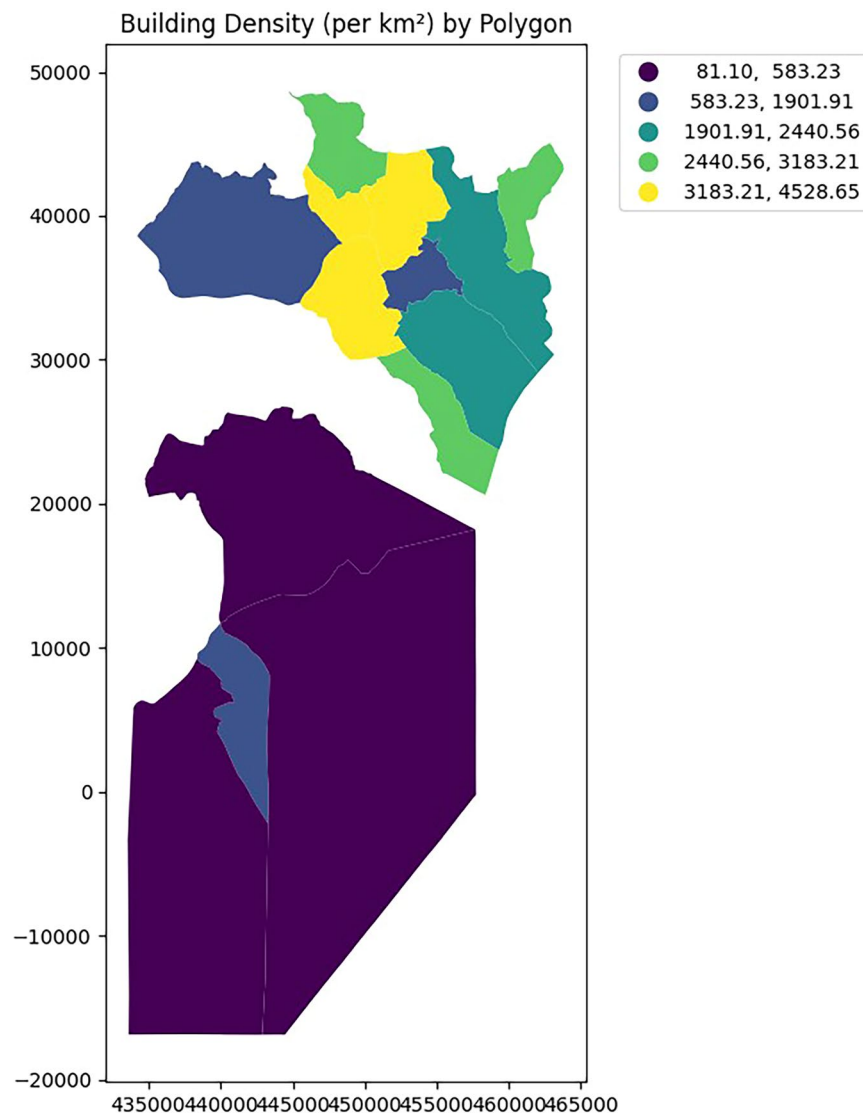


Fig. 3 Building density (per km²) across administrative polygons in Kampala and Entebbe. This map visualizes polygon-level building density to characterize broad urban morphology patterns. Colours represent density ranges from low (approximately 81 buildings/km²) to high (approximately 4528 buildings/km²), highlighting the spatial contrast between dense mixed-use urban cores and low-intensity residential or peri-urban areas. These structural characteristics influence noise propagation and environmental sound levels.

audio samples. Field teams were encouraged to move throughout their assigned areas to ensure broad geographic representation and reduce spatial clustering in the data.

Throughout the data collection period, submissions were continuously monitored via the ODK Central dashboard. Additional quality assurance was performed by programmatically downloading the data and metadata using the ODK API, followed by analysis in Python with the pandas library. Random samples of audio were manually reviewed to verify data quality and accuracy of sound categorization.

Urban morphology metrics and density surfaces. We characterized the urban form of Kampala and Entebbe using complementary morphology descriptors that capture both coarse administrative patterns and fine-scale built intensity around each recording location.

Building density (polygon level). We aggregated building points to administrative polygons and computed building density as buildings per km². This summarizes the spatial hierarchy of centers and corridors across each city and highlights contrasts between dense mixed-use cores and lower-intensity residential tracts (Fig. 3).

Continuous building density surface. To capture neighborhood-scale structure relevant to acoustic propagation, we constructed a continuous density surface from building points using a radial count normalized by the search area. For each 50 m raster cell we counted buildings within a radius $R = 300$ m and divided by πR^2 ,

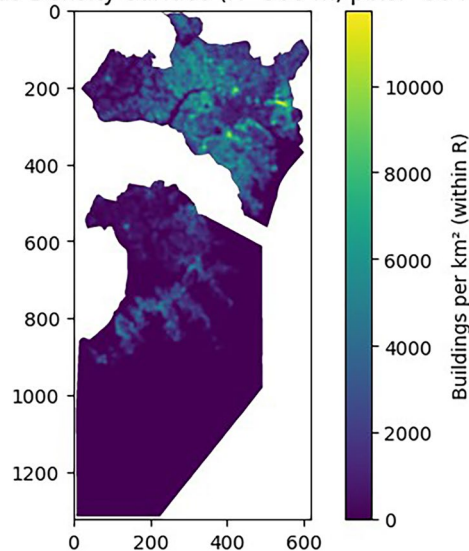
Continuous Density Surface ($R=300$ m, pixel=50 m)

Fig. 4 Continuous neighbourhood-scale building density surface ($R = 300$ m, pixel = 50 m). This raster representation shows the fine-scale spatial distribution of building density around each grid cell. Density values represent the number of buildings within a 300 m radius, normalized per km^2 . The surface reveals corridor patterns, street-canyon effects, and mixed-use clusters that polygon averages cannot capture. This morphology metric is used to contextualize ambient noise levels in the study.

yielding buildings per km^2 (Fig. 4). This representation reveals street-canyon and corridor effects that polygon averages cannot show.

Network and topography. We derived standard network metrics in the R -radius neighborhood of each sample, including road density (km per km^2) and intersection count, and extracted mean local slope from a DEM. These proxy traffic intensity, congregation nodes, and topographic constraints on both built form and sound propagation.

Sample-level extraction. Morphology metrics were sampled at each audio location or within a circular buffer of 100 m to reflect the near-field environment in which most urban sources and barriers operate. All metrics were aligned to the same spatial reference to ensure consistent neighborhood definitions across Kampala and Entebbe.

Post-processing and Privacy. To ensure both data quality and compliance with privacy standards, a series of post-processing steps were applied to the collected noise data. First, all raw audio samples were subjected to voice activity detection (VAD) using an open-source deep learning model. This process was used to automatically detect the presence of human speech. The analysis confirmed that the recorded samples did not contain intelligible or private conversations.

Duplicate entries and corrupted files were identified and removed through automated checks and manual inspection. Duplicates were detected using hash-based matching on audio content and metadata such as timestamps and geolocation. Corrupted files, identified by invalid audio headers or unreadable formats, were discarded to maintain the integrity of the dataset.

These post-processing procedures helped create a high-quality, ethically sound dataset suitable for urban environmental monitoring, research, and planning applications.

Categories of Noise Collected. Data collection covered 19 distinct noise categories. These categories were defined in consultation with the National Environmental Management Authority (NEMA), which is the national regulatory agency responsible for environmental oversight in Uganda, together with the Kampala Capital City Authority (KCCA) and the Entebbe Municipal Council.

The process was informed by community noise complaints, which typically arose when noise sources were perceived as disruptive. In response to such complaints, regulatory officers from these bodies were deployed at different times of day and night to perform ground-truth measurements using noise decibel meters.

These inspections were intended to determine whether noise levels exceeded permissible thresholds set out in *The National Environment (Noise Standards and Control) Regulations, 2003* (enacted under sections 28 and 107 of the National Environment Act, Cap. 153)^{18–24}, and, where necessary, to guide subsequent warnings or enforcement actions.

From this community-driven reporting and regulator verification process, the following categories of noise were identified as being of primary concern: animal, bar-restaurant-night-club, bodaboda-motorcycle, car-alarm, car-or-truck, community-radio, construction-site, crowd-noise, fabrication-workshop, generators, hawker-vendor, herbalists, mobile-music, motor-vehicle-horn, motor-vehicle-siren, religious-venue, schools, schools and other.

Part	
Name	Description
animal	Any animal noise
bar-restaurant-night-club	Music, people talking at a bar/restaurant/club
bodaboda-motorcycle	Motorcycle moving on the road, engine ignition
car-alarm	Car alarm
car-or-truck	vehicles excluding motorcycles
community-radio	Mega phones
construction-site	Drilling, hammering etc at a construction site
crowd-noise	Noise from large groups of people
fabrication-workshop	Metal fabrication workshops, welding
generators	Generator running
hawker-vendor	Hawker/vendor advertising or selling their goods
herbalists	People advertising and selling herbs and any other local drugs
mobile-music	Cars advertising events, concerts, campaigns
motor-vehicle-horn	Vehicle hooting / honking
motor-vehicle-siren	Siren from vehicles such as police and ambulance
religious-venue	Any religious site
schools	Ambient noise from schools
street-preachers	Person preaching on the street
others	Any other noise

Table 2. Categories table.

Data Records

The dataset has been deposited on Figshare and is available at <https://doi.org/10.6084/m9.figshare.301689014>.

It consists of audio recordings with detailed metadata. Each recording is represented as a single data point, where the raw waveform is linked to a comprehensive set of descriptive variables. The dataset is designed to support research in noise classification, environmental sound analysis, and geospatial noise monitoring (Table 2).

File Structure. The dataset is organized into a clear and reproducible directory layout:

```
urban-noise/
  audio_files/
    0a0b72efcfe44acb.ogg
    0a0c26863e4b423f.ogg
    0a0ce5ea12ae4b56.ogg
    ...
  metadata.csv
```

Variables and Column Headings. The `metadata.csv` file describes each audio recording in the dataset. Each row corresponds to one `.ogg` file and provides the following fields:

- **filename:** String. The name of the audio file (e.g., `7f961a5e3df14933.ogg`).
- **class:** String. The noise category of the audio file (e.g., `'car-or-truck'`, `'crowd-noise'`).
- **class_id:** Integer. A numeric identifier for the `class` category.
- **noise_measurement:** Float. The measured noise level at the time of recording, in decibels (dB).
- **latitude:** Float. The geographical latitude of the recording location (decimal degrees).
- **longitude:** Float. The geographical longitude of the recording location (decimal degrees).
- **altitude:** Float. The altitude of the recording location, in meters above sea level.
- **accuracy:** Float. The precision of the recorded geographical coordinates, in meters.
- **submitter_id:** Integer. A unique identifier for the contributing device or submitter.
- **region:** String. The administrative region in Uganda where the recording took place (e.g., `'Kampala'`).

Data Overview. The dataset is self-describing and designed for reproducibility. The combination of raw audio files and rich contextual metadata allows researchers to explore spatial, temporal, and categorical dimensions of urban noise in Uganda.

The availability of both CSV and Parquet metadata formats supports efficient large-scale processing. Together, these attributes ensure the dataset is suitable for applications in machine learning, policy-driven noise mapping, and environmental monitoring.

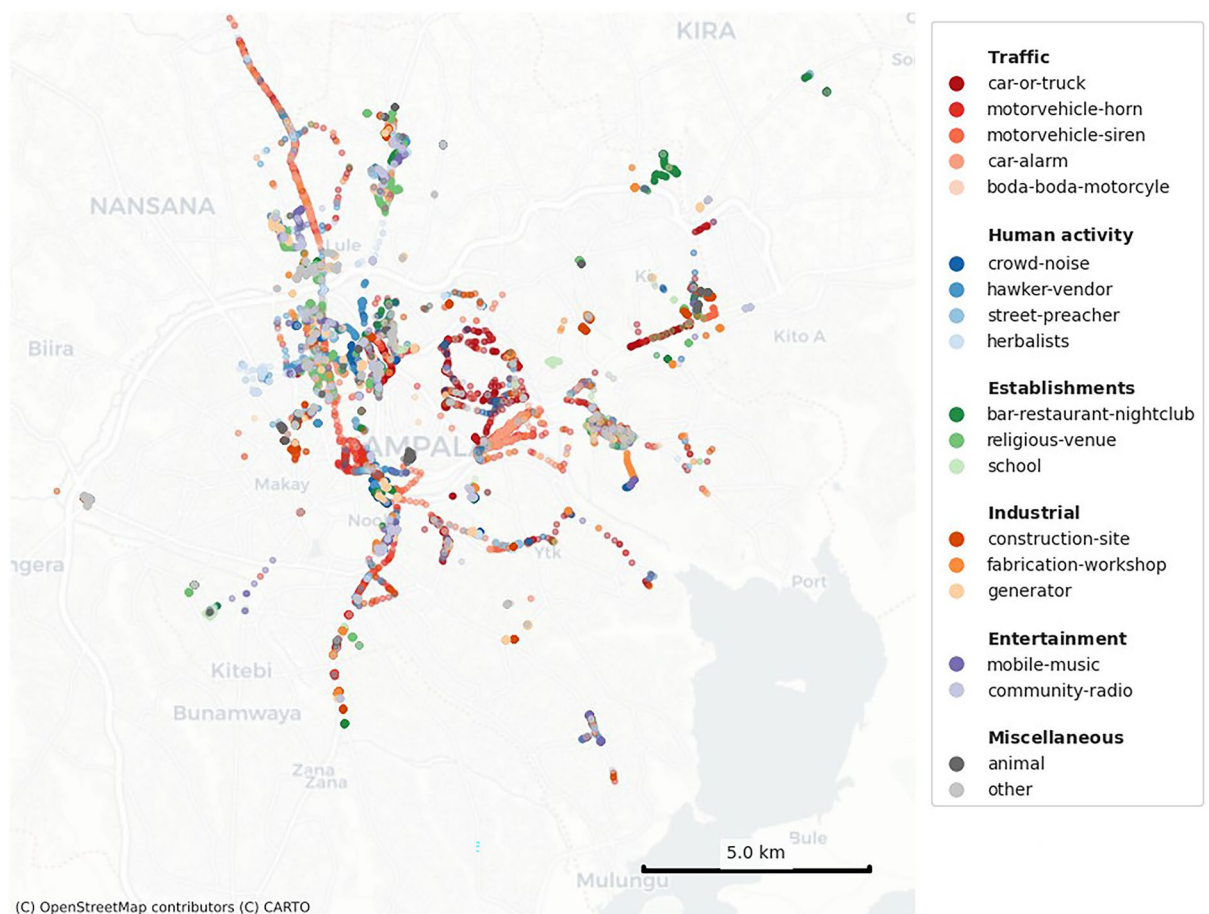


Fig. 5 Spatial distribution of noise categories recorded across Kampala City. This map displays category-level labels for environmental noise samples collected throughout Kampala. Colours correspond to thematic groups including traffic (e.g., car-or-truck, boda-boda-motorcycle), human activity (crowd noise, street preachers), establishments (bar-restaurant-night-club, religious venues), industrial sources, entertainment, and miscellaneous classes. The visualization illustrates broad spatial coverage and category diversity across major transport corridors and commercial zones.

Geographical coverage and category distribution. The dataset includes labeled recordings across Kampala and Entebbe, enabling descriptive visualization of where categories were observed and where monitoring took place.

Category coverage. Figs. 5 and 6 depict the spatial distribution of recordings by label category. These maps summarize where traffic-related, commercial, religious, construction, and other classes were observed and illustrate broad coverage across major corridors and activity centers.

Monitoring footprint. Fig. 7 shows the geographical distribution of all monitoring points. Dense clusters appear in central Kampala and Entebbe, with additional coverage along major access routes. These figures are descriptive of dataset coverage rather than analytical results.

Technical Validation

File-level signal checks.

- **Decodability & integrity:** We batch-decoded all .ogg files and verified read success, channel count, sample rate, and duration.
- **Silence & dropouts:** We measured frame-wise RMS and flagged recordings with median RMS below a low-energy threshold (suggestive of mic failure or disconnection). Long flat-line segments were inspected and removed if due to device faults.
- **Spectral sanity:** We summarized band-energy in low/mid/high ranges (e.g., 0-250 Hz, 250-2 kHz, 2-8 kHz) to detect abnormal spectra (e.g., DC offsets, aliasing). Outliers (> 3 MAD from median within a session/device) were reviewed.

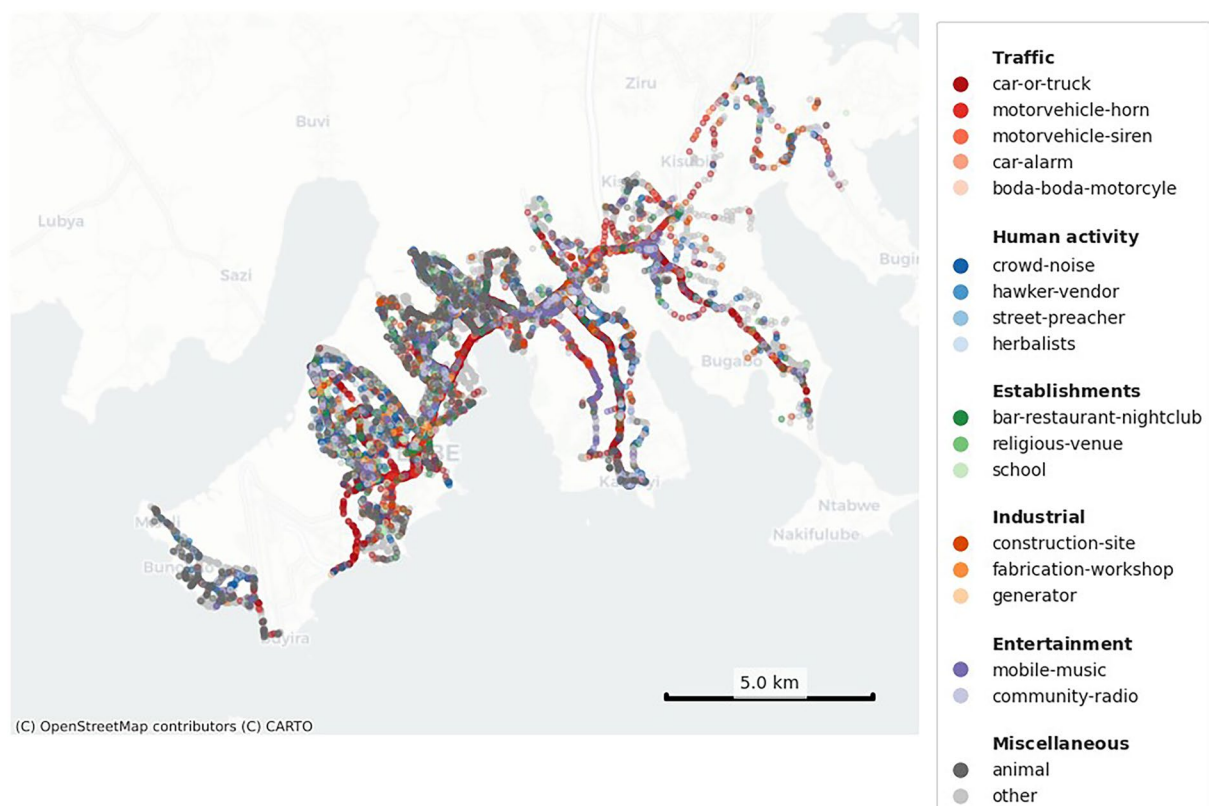


Fig. 6 Spatial distribution of noise categories recorded across Entebbe Municipality. This map shows the spatial footprint and distribution of labeled noise recordings within Entebbe. Similar to Kampala, the categories include traffic, human activity, establishments, industrial sources, entertainment, and miscellaneous noises. Dense clusters appear along key transport routes and activity centres, reflecting the city's unique shoreline- and airport-driven urban form.

- **Duplication:** Exact duplicates were identified via file hash (MD5/SHA-256). Near-duplicates were detected by audio fingerprinting (Chromaprint) and by comparing duration ± 10 ms and mean RMS; confirmed duplicates were removed.

Usage Notes

Limitations and challenges. The dataset collection was subject to some limitations, in particular:

- The risk of phone theft as the data collectors moved around the city, meaning fewer recordings at night-times and from particular urban locations.
- Rain was a major hindrance to the data collectors' movements on some days.
- Duplicate noise samples submitted by some of the data collectors.

Example SPL mapping. As an example of downstream use, Fig. 8 visualizes mean SPL across Entebbe from April 5 to May 17, 2023 (24,150 measurements). Such surfaces are derived from the dataset and illustrate how users can generate local exposure maps for planning or public communication.

Hourly and diurnal summaries. Figure 9 provides a descriptive hourly summary (for example, median and interquartile range across a 24-hour cycle) of SPL across the dataset. Daytime hours typically show higher medians (often 45–50 dB), with early mornings quieter. Peaks around 21:00–22:00 reflect localized evening activity patterns and can be reproduced by users for different areas or time windows.

Day vs. night comparison. Figure 10 contrasts daytime and nighttime SPL distributions, illustrating typical diurnal separation in urban environments. Users can adapt these summaries by subsetting regions, categories, or periods of interest.

Morphology-aware interpretation. *Urban morphology as a determinant of noise.* The density and configuration of the built environment are key correlates of ambient noise in Kampala and Entebbe. Continuous

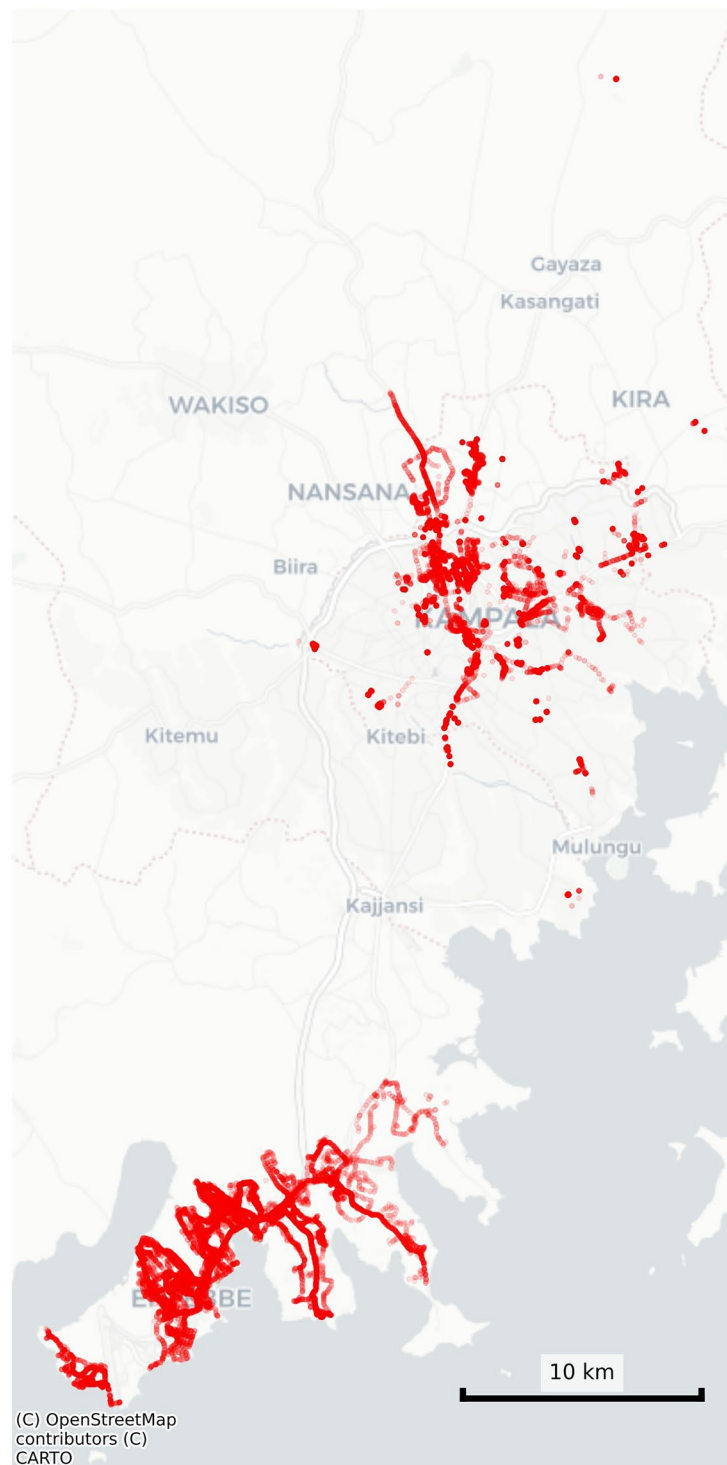


Fig. 7 Geographical coverage of environmental noise monitoring points in Kampala and Entebbe. This figure shows the complete set of 61,821 noise measurement locations collected by the field teams. Points indicate individual recording positions, demonstrating extensive spatial coverage across both cities, including major roads, commercial centres, residential neighbourhoods, and transportation corridors.

building density and street-network intensity (road and intersection density) are associated with higher SPL, whereas topographic steepness modestly attenuates levels. These relationships substantiate that the spatial structure of each city, including the polycentric, corridor-driven layout of Kampala and the shoreline- and airport-constrained layout of Entebbe, shapes the acoustic landscape. This observation supports targeted interventions (for example, enforcement and façade or setback guidelines along dense corridors and at high-intersection nodes) and provides context for interpreting our time-of-day and regional analyses.

Overall noise levels in Entebbe City

May 2023

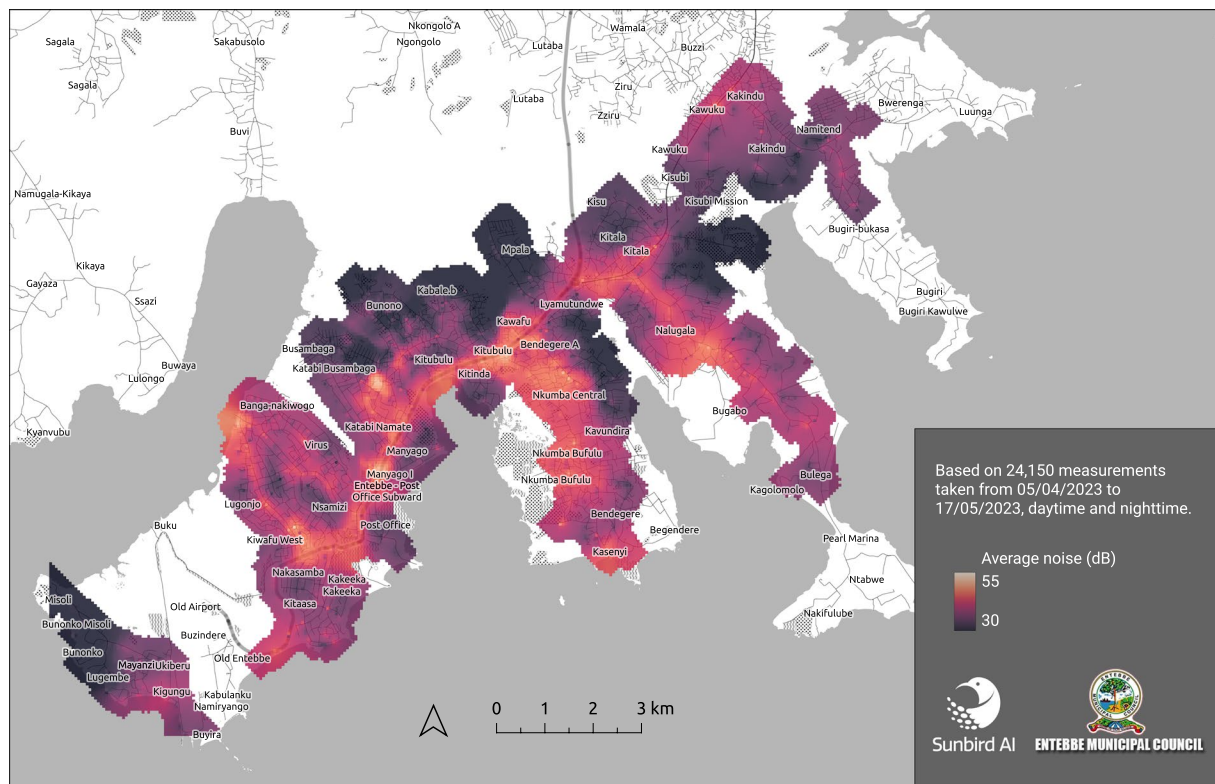


Fig. 8 Spatial distribution of average noise levels (dB) in Entebbe City for April–May 2023. This heatmap visualizes mean sound pressure levels derived from 24,150 measurements collected between 5 April and 17 May 2023. Higher noise intensities are observed along busy corridors, market areas, and near the airport, while quieter zones appear in residential and peri-urban areas. The map demonstrates how the dataset supports noise exposure mapping for urban planning and public health applications.

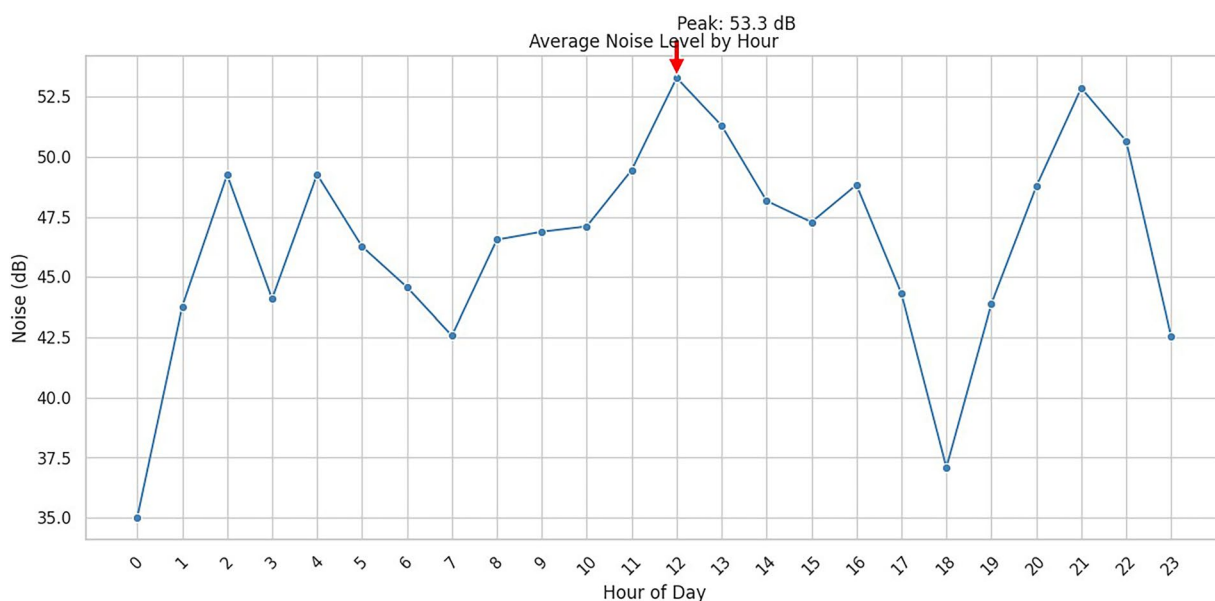


Fig. 9 Hourly variation in average sound pressure levels across the dataset. This line plot summarizes the typical 24-hour noise cycle using hourly median SPL values. Daytime hours generally exhibit higher noise levels (often around 45–50 dB), with peaks around late afternoon and evening (notably around 12:00–14:00 and 21:00–22:00). Early morning hours are relatively quieter. This figure highlights temporal patterns in urban soundscapes across Kampala and Entebbe.

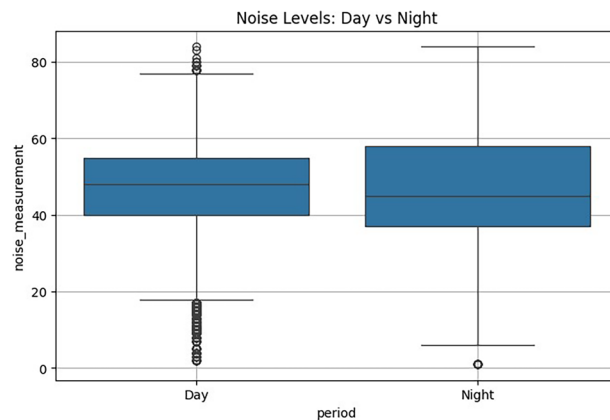


Fig. 10 Comparison of daytime and nighttime sound pressure level distributions. This boxplot contrasts SPL measurements collected during daytime and nighttime periods. Both periods show wide variability, but daytime levels trend slightly higher due to increased traffic and human activity. Nighttime distributions retain notable upper-tail noise events linked to nightlife, transportation, and periodic disturbances. This distinction supports diurnal noise analysis and policy planning.

Positioning relative to existing datasets. The Urban-Noise-Uganda-61k resource includes classes and geospatial detail (labels, timestamps, coordinates, SPL) that are often absent in other collections. Coverage of context-specific categories (e.g., religious venues, street preachers, hawkers/vendors, community radios, mobile music, nightlife venues) increases relevance for low-resource urban settings and supports transfer to cities with similar infrastructure and activity patterns.

Data availability

All data underlying this study are available in the Urban-Noise-Uganda-61k dataset hosted on Figshare (<https://doi.org/10.6084/m9.figshare.30168901>)⁴. The repository contains the raw audio recordings in .ogg format together with accompanying metadata files (CSV).

Code availability

Example code for accessing and using the Urban Noise Uganda 61K dataset is available on Hugging Face: <https://huggingface.co/datasets/Sunbird/urban-noise-uganda-61k>. This repository includes scripts for loading the audio files and metadata.

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Author contributions

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Competing interests

The authors declare no competing interests.

Additional information

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